01

DATALENS

Dataset quality, without the upload

A browser-based profiler that turns raw files into evidence, charts and reproducible cleaning SQL.

DUCKDB-WASM / REACT / ECHARTS

02

THE PROBLEM: UNTRUSTED INPUT FILES

Hidden quality issues

Missing values and duplicate rows can distort an analysis before a dashboard even exists.

Repeated manual checks

One-off notebooks scatter profiling logic and make the next file harder to compare.

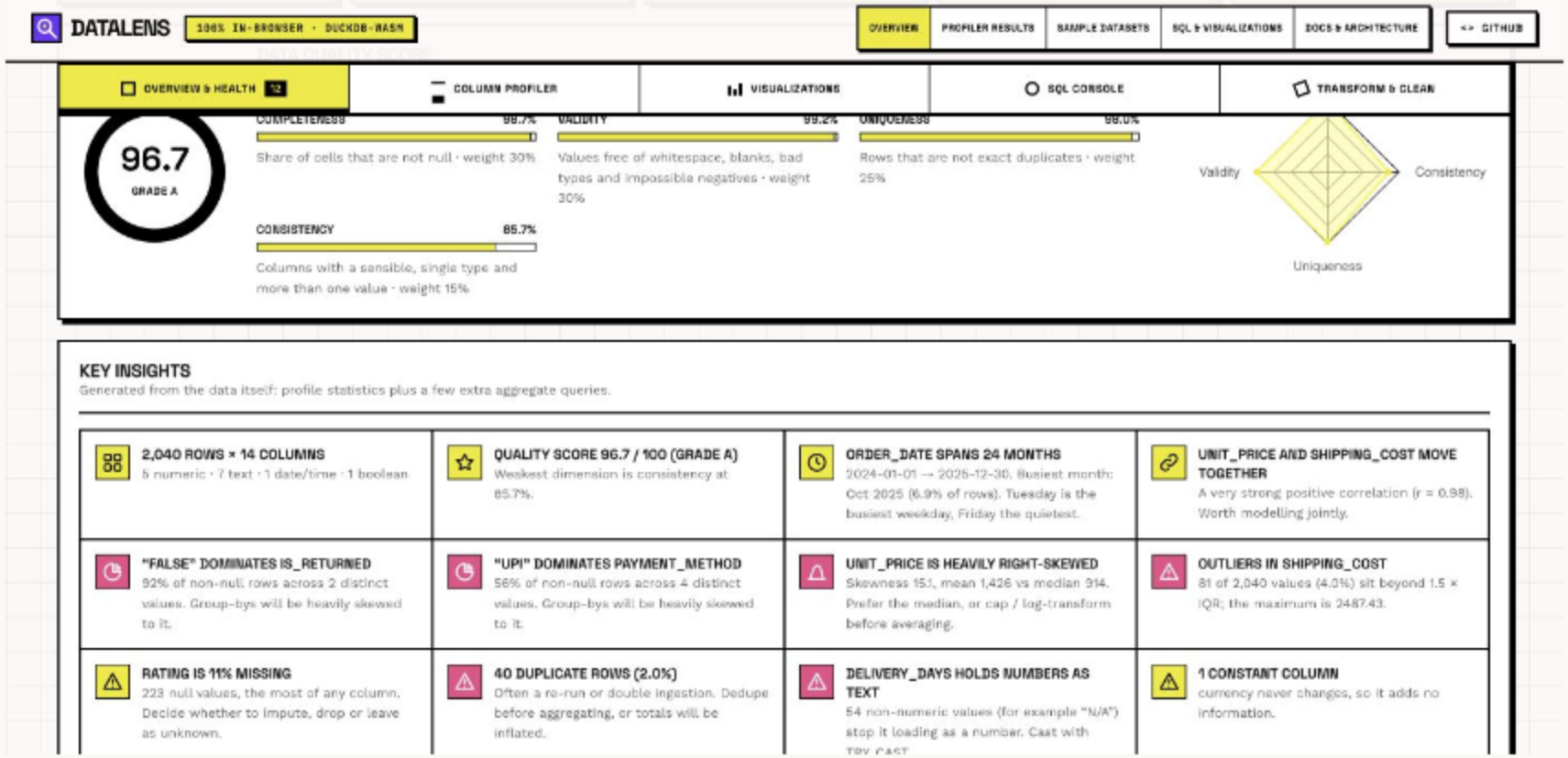
Sensitive datasets

Uploading source records creates a privacy barrier for quick exploratory analysis.

The response: run profiling and cleaning inside the browser.

03

THE PROFILE EXPLAINS ITS SCORE



96.7 / 100

Synthetic orders sample

A high overall grade can still hide issues that matter.

40 duplicate rows
223 missing ratings
129 untrimmed city values

04

HOW I BUILT THE PROFILING ENGINE

01 LOAD Register the local file in DuckDB-WASM. Keep source rows in the browser.

02 COMPUTE Use SQL passes for wide aggregates, IQR outliers and exact duplicates.

03 EXPLAIN Weight completeness 30%, validity 30%, uniqueness 25%, consistency 15%.

04 REPRODUCE Generate one cleaning statement, reprofile the result and export the data.

05

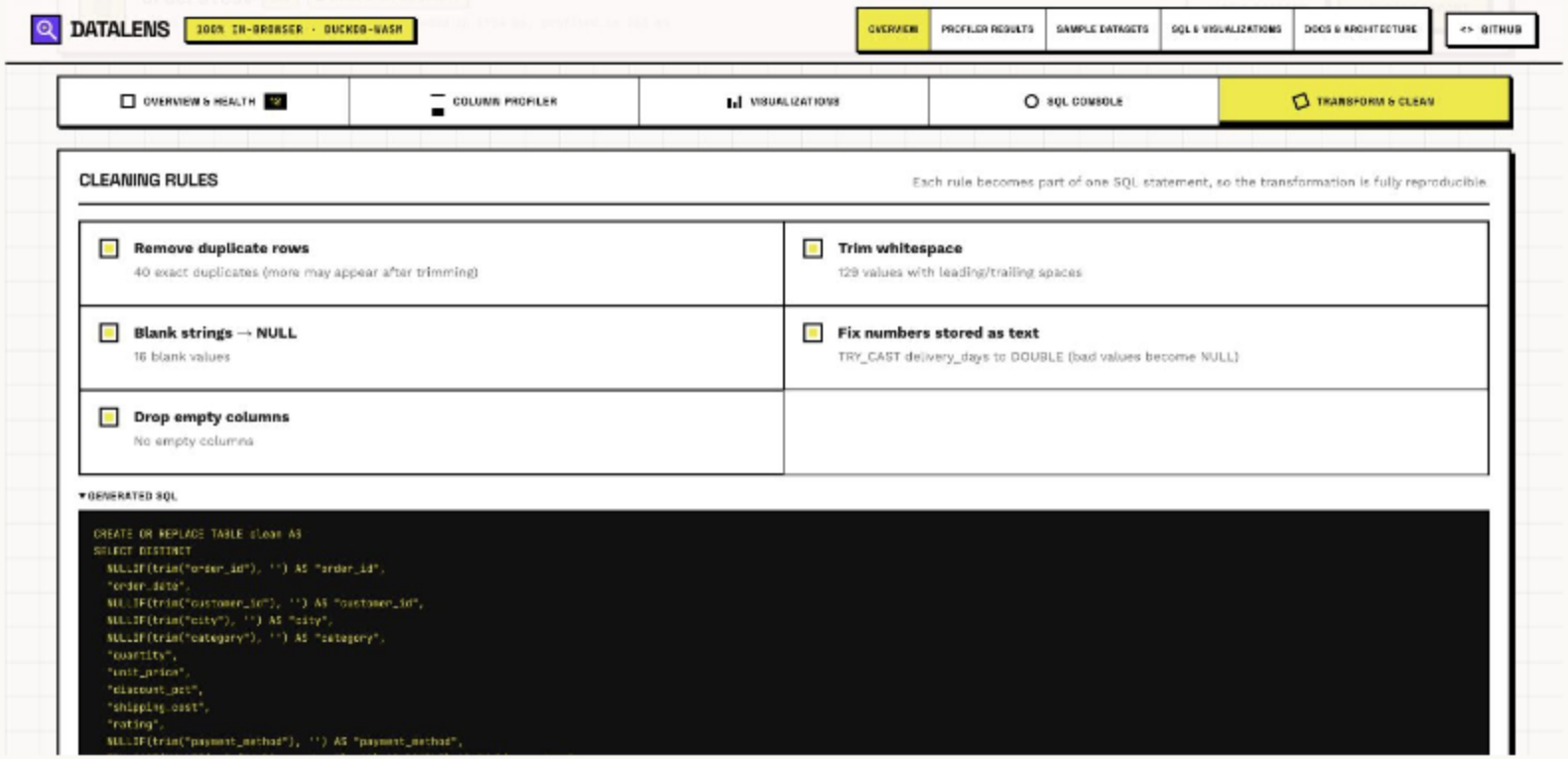
CHARTS TURN STATISTICS INTO QUESTIONS



Synthetic demo: monthly patterns reveal when to investigate, not why a change occurred.

06

CLEANING THAT LEAVES AN AUDIT TRAIL



Readable SQL

Dedupe, trim strings, normalize blanks and fix types with a reproducible transformation.

Boundaries

Browser memory limits file size. Some readers download DuckDB extensions. Demo results are synthetic.

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